The Long-Run Effects of Parental Homeownership and Housing Quality on Later Life Outcomes

by

Garret Christensen
FDIC

John Voorheis
US Census Bureau

CES 26-54 **August 2026**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

Papers in the series are posted on www.census.gov/library/working-papers/series/ces-wp.html. For information about the series, contact Danielle H. Sandler, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov.

Abstract

We measure the effects of housing quality and living in renter- versus owner-occupied housing as a child on later life outcomes. Homeownership is the primary source of most US households' wealth, and homeownership and housing quality potentially have large impacts on children, yet we know relatively little about any long-term effects. Ownership and housing quality are correlated with income, neighborhood characteristics, and many other socioeconomic characteristics, making causal estimation difficult. Using the US Census Bureau's data linkage infrastructure, we create a new data linkage between the American Housing Survey (1995-2002) and administrative tax data (2016-2019), to investigate adult outcomes such as earnings and having a mortgage or owning a home, while comprehensively controlling for a wide variety of variables, including housing quality. We find strong positive correlations with growing up in a parent-owned home, approximately $2,000 higher incomes and a 6 percentage point increased likelihood of ownership. However, we find no evidence of causal effects: unobservables that are significantly less correlated with outcome and treatment than observables would drive the coefficients of interest to zero, and family fixed effects models that compare siblings to each other show no effect of years of childhood spent in an owned home or being born into an owned home.

[*] Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7505723: CBDRB-FY23-CES019-006, CBDRB-FY23-CES019-012, CBDRB-FY23-CES019-021, CBDRBFY25-CES019-005, CBDRB-FY25-CES023-018). The authors have no funding sources, financial relationships, nor conflicts of interest to disclose.

1 Introduction

Some believe that homeownership leads to large societal benefits. As part of his push for an "ownership society," President George W. Bush said "...if you own something, you have a vital stake in the future of our country. The more ownership there is in America, the more vitality there is in America, and the more people have a vital stake in the future of this country."[1] As a policy matter, homeownership is heavily subsidized in the United States. The Treasury Department reported the exclusion of imputed rental income and the mortgage interest deduction as their second and seventh largest tax expenditures, roughly $130 billion and $30 billion, respectively.[2] The majority of mortgage interest deductions are taken by households earning $100,000 or more[3]. Homeownership and housing quality potentially have large impacts on children, which could justify these public expenditures, yet we know relatively little about any long-term effects. Further, ownership and housing quality are correlated with income, neighborhood characteristics, and many other socioeconomic characteristics, making causal estimation difficult. Using Census and IRS administrative data, we comprehensively control for a wide variety of variables, including household and county characteristics, to investigate adult outcomes such as earnings and having a mortgage or owning a home.

Homeownership could lead to more stable residency and fewer schooling changes or disruptions, thus leading to improved educational outcomes [Green and White, 1997, Haurin et al., 2002, Bramley and Kofi Karley, 2007, Mohanty and Raut, 2009], though research with clearer identification strategies calls this hypothesis into question [Barker and Miller, 2009]. Poor quality housing could lead to negative health outcomes for children or adults [Harville and Rabito, 2018, Boch et al., 2020]. Homeownership can also be a savings mechanism, leading to intergenerational transmission of wealth [Benetton et al., 2022]. We are the first US-focused paper on homeownership that we are aware of to use tax records to accurately measure income, and investigate the causal effects, but unfortunately our causal estimates are not able to capture savings or wealth mechanisms, due to a lack of data.

We have linked data on children from the 1995 through 2002 American Housing Surveys (AHS) to their recent tax returns in the Census FSRDC environment. That is, their parents responded to the AHS,

[1] See https://georgewbush-whitehouse.archives.gov/news/releases/2004/08/20040809-9.html

[2] See https://home.treasury.gov/policy-issues/tax-policy/tax-expenditures.

[3] See https://taxfoundation.org/research/all/federal/who-benefits-home-mortgage-interest-deduction/ and https://www.cbpp.org/research/mortgage-interest-deduction-is-ripe-for-reform.

and described the home in great detail as well as basics of family structure and income. Children in those families are the main focus of our study: we use data from their recent tax returns as outcome measures. To clarify terminology, we use 'parental homeownership' to describe whether the parents owned a home in 1995 through 2002 when the subject was a child. 'Young adult homeownership' and 'owning a home as a (young) adult' refer to more recent (2016-2019) data when the subject is older and owns a house later in life. We control for parental income, education, demographics, neighborhood characteristics, important housing attributes such as crowding (the occupants per room) and housing quality, using a multidimensional index of housing issues such as plumbing, heating, electrical, and paint. All these characteristics are measured when the subject was a child.

Our linked survey and administrative records strategy yields a sample size over 100,000 observations. We find large and strongly statistically significant positive associations between growing up in parent-owned housing and owning a home or having a mortgage as a young adult, on the order of a 6 percentage point increase on a base of roughly 35%. The same strong positive association holds for income as well: incomes are roughly $2,000 higher for children of owners. Interactions show the positive effects are smaller for some minorities than for whites, consistent with the idea that minorities are less able to accrue and pass on housing wealth (or high incomes) to their children. (See Kermani and Wong [2024], Diamond and Diamond [2024] and Binder et al. [2026] for discussion of returns to housing by race.)

We find little evidence that the positive associations are necessarily causal. Unobservable variables such as a family's innate desire to own a home, tradition of owning, or accumulated wealth could be the real driver of homeownership and thus bias our estimates. To attempt to address this concern, we use three methods. First, we use the Oster [2019] method of partial selection correction to put lower bounds on the estimates from our simple linear regressions[4]. Second, we estimate within-family models that compare siblings within a given family and flexibly control for any unobservable family characteristics that are constant across siblings. Finally, we estimate within-family-over-time models that add controls for changes to major family characteristics such as income and education, affecting siblings within a family at different ages.

These three methods produce a consistent picture of no evidence of causal effects of parental homeownership, particularly when comparing between siblings in the same family. That is, unlike Chetty and Hendren

[4]Interested readers should also see Diegert et al. [2022] which discusses a method relaxing the exogeneity assumptions about unobservables that is used in Oster's method.

[2018] that shows a clear linear dose-response-like effect for each year of living in a better neighborhood, we find no such effect for parental homeownership.

The rest of the paper proceeds as follows: we discuss the background of homeownership and the related literature in section 2. We describe our data sources in section 3. We provide our results in section 4, and conclude in section 5.

2 Background and Literature

The majority of American households own their home, and homeownership is one of the largest sources of wealth for American households[5]. Yet there are significant differences in the homeownership rate across racial groups which have been stubbornly persistent over the past several decades.

Figure 1 shows the fraction of households in AHS data that report to own their home (with or without a mortgage) by race and ethnicity. One can see that homeownership decreased during the great financial crisis, and more so for minorities than for whites. The situation has improved over the decades for Asian households, but the gap for Blacks and Hispanics of any race has not decreased much in thirty years; if anything it increased during the great financial crisis. Figure 2 shows the trend for crowding, the measure of whether a household has more than 1.5 people per bedroom. The rate of crowding has decreased for all groups, and the relative ranking by race and ethnicity has remained unchanged.

[5]See `https://www.census.gov/content/dam/Census/library/publications/2023/demo/p70br-183.pdf`. According to the Census' 2021 Survey of Income and Program Participation, retirement accounts have now become a slightly larger source of wealth than housing equity.

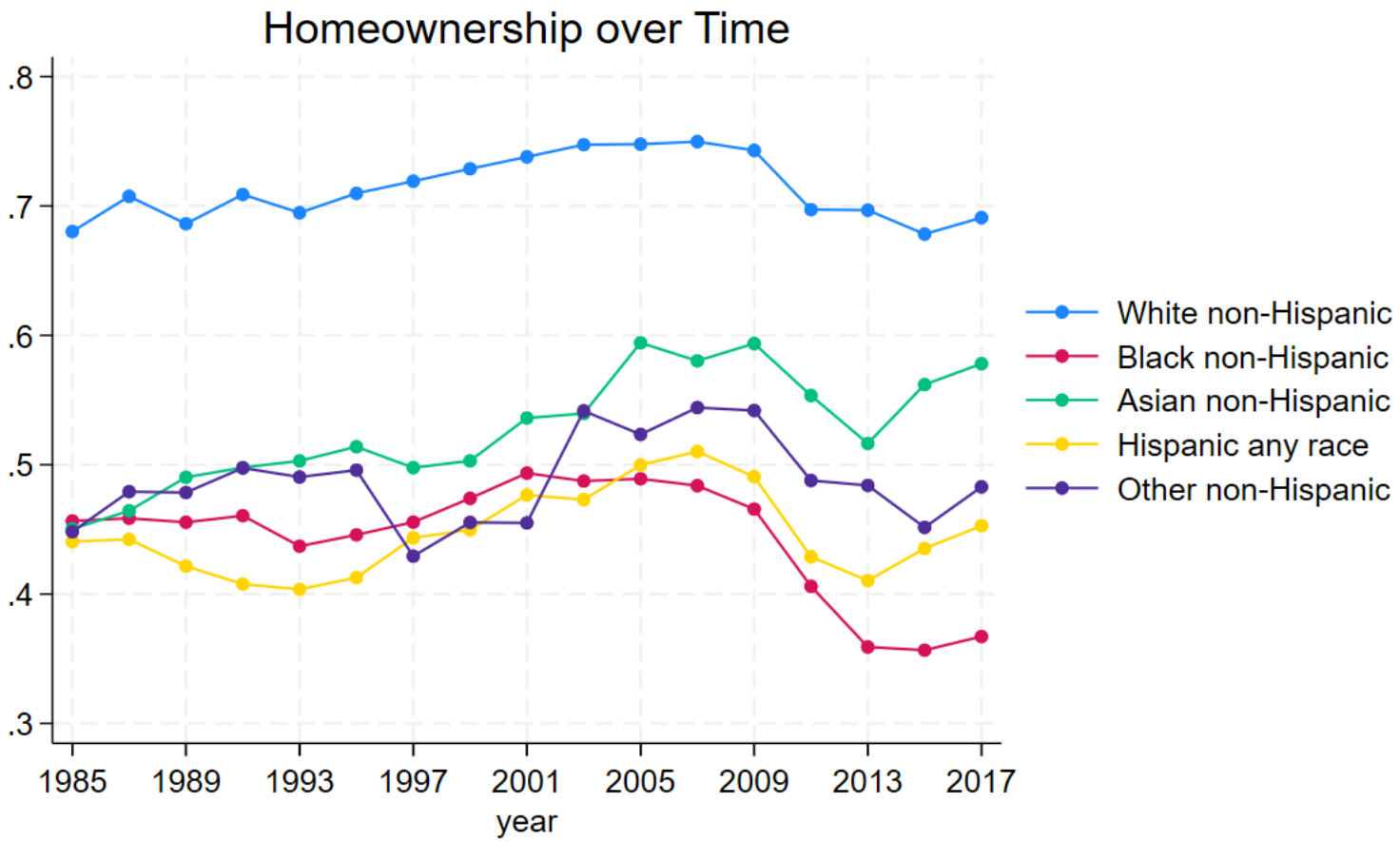


Figure 1: **Homeownership Rates over Time** This chart shows the fraction of homeownership by race and ethnicity using AHS data.

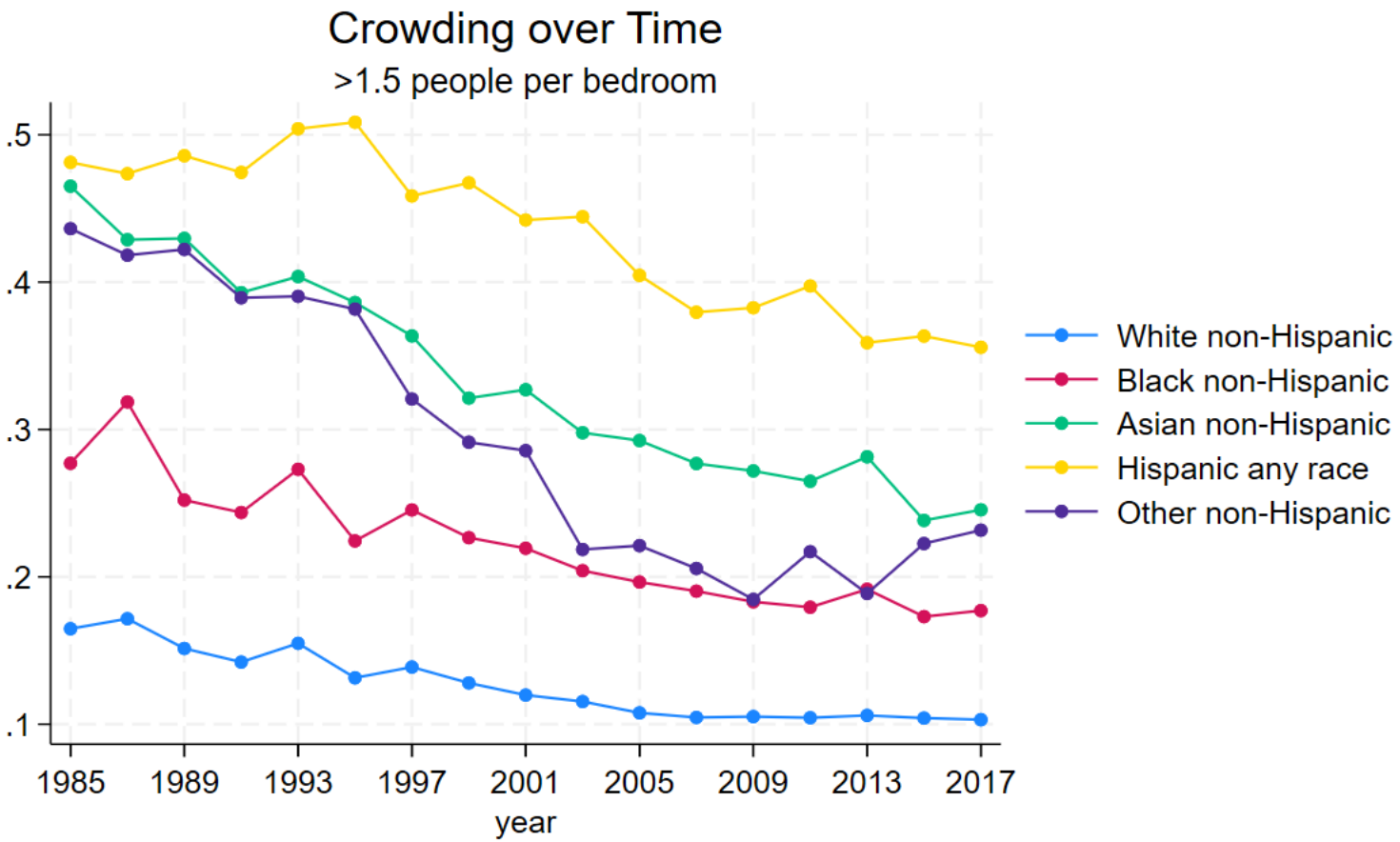


Figure 2: **Crowding Rates over Time** This chart shows the fraction of crowding by race and ethnicity using AHS data.

There is a limited amount of previous work attempting to identify causal estimates of parental homeownership. HUD's *Cityscapes* featured a few articles on the question of "Do Kids of Homeowners Do Better Than Kids of Renters?" but opinions differed and causality was unclear [Green, 2013, Haurin, 2013, Barker, 2013, Newman and Holupka, 2013]. One of the best pieces of causal evidence is Benetton et al. [2022], which uses credit bureau panel data and an IV strategy to show that parental wealth extraction from homes increases the likelihood of child homeownership as a young adult. Children of parents who own are 1.4 percentage points (28%) more likely to own at age 25. The age limitation is due to the limited duration of the credit bureau data (roughly 7 years), and is relevant, since the homeownership rate at age 25 is low. Blickle and Brown [2019] uses Swiss panel data on 5,000 households to show that receiving a wealth transfer (not necessarily due to parental homeownership) are associated with a 6 to 8 percentage point higher likelihood of transition to ownership. They find this is likely due to relaxation of borrowing constraints rather than wealth effects, and they find no similar inducement of owners to trade up to larger homes.

Other papers on housing wealth include Charles and Hurst [2002], which shows that the primary source of the black-white gap in transition to homeownership is that blacks are less likely to apply for mortgages, and finds an association between parental wealth and child homeownership using 1,500 child-parent pairs from the Panel Survey of Income Dynamics (PSID). Guiso and Jappelli [2002] show similar results in Italy based on gifts and bequests. Blanden and Machin [2017] and Blanden et al. [2023] document persistence in the rate of homeownership in the UK, and Englund et al. [2014] document the same in Sweden and show that the intergenerational persistence of wealth comes mostly through the housing channel. Binder et al. [2026] leverages linked survey and administrative records to study the intensive margin of the intergenerational transmission of housing wealth, though only descriptively.

In addition to wealth, homeownership may increase residential stability and thus improve schooling outcomes. Mobility rates are significantly higher for renters than owners (roughly 40% compared to 10%, according to the US Census Bureau[6].) Higher mobility is often linked to declines in childhood schooling outcomes [Government Accountability Office, 2010]. However, much of the research on school changes fails to account for different reasons for moves. Schwartz et al. [2017] differentiates between structural (boundary changes, sale by the owner of a building renters live in) and nonstructural (other, such as changes to family

[6]See `https://www.census.gov/data/tables/time-series/dec/coh-movers.html` and `https://www.census.gov/library/stories/2017/12/lower-moving-rate.html`.

structure) and find that while structural moves hurt school performance, nonstructural moves actually lead to improved school performance. Thus it is not entirely clear what effects we should assume homeownership (and fewer school changes) would have.

Other papers looking at housing quality use a binary measure of "inadequacy," that applies to only 5% of the housing stock. Instead we use a 60-variable index from Eggers and Moumen [2013] to create a continuous quality measure (hereafter Poor Quality Index, PQI). An additional aspect of housing quality that is potentially important is that of lead paint, which is in an estimated 25% of homes [Jacobs et al., 2002]; we proxy for this using date of construction and household reported paint issues. These quality aspects are important because they can have well-documented long-term effects on health, and likely income and thus homeownership as an adult. For example, there is no safe level of lead (present in paint until 1978) in the human body, and research has shown effects to be very large, especially in children. (See for example Aizer et al. [2018], Aizer and Currie [2019].)

3 Data

We use three primary datasets: the American Housing Survey (AHS), IRS tax records, and Black Knight property tax records. We also add home price data from the Federal Housing Finance Agency (FHFA).

The AHS began in 1973 and has been conducted annually or biennially since. The survey asks numerous questions about the physical and financial characteristics of the home as well as its inhabitants. At the national level it is a structure-based longitudinal survey (meaning the survey returns to the same building year after year regardless of who occupies it). We use the data from 1995-2002, which consists of the 1995, 1997, 1999, and 2001 national surveys, and the 1996, 1998, and 2002 metropolitan sample—the metro sample is not longitudinal, but rather a snapshot from different metro areas in each year. The Census Bureau has built crosswalks for the AHS for data from 1995 to the present using the person identification validation system (PVS), which matches Protected Identification Key (PIK) numbers for each person (and Master Address File ID (MAFID) for each address) in any of the many datasets available in the Census Bureau's secure computing environment. We restrict the sample to non-heads of household under 25 years old the first time they appear in the data, and to those coded as children or grandchildren of one of the families in

the household. 190,000 of the 220,000 observations of children in the AHS are successfully assigned PIKs.

The IRS tax data includes income measures from W-2 and 1040 forms as well as information returns (forms 1099 and 1098) filed by third parties. We use data from the years 2016-2019. Sample children are aged 15 to 47 in the outcome data, though we restrict the regression sample to those 21 or older and those who have filed taxes (not as a dependent) at least once. The Census Bureau's version of these data has indicators for each individual schedule filed, but not contents of those forms. That is, we can observe whether a filer itemized their deductions, but have no information on details such as property or state tax payments. Mortgage interest returns (form 1098) are of particular interest to us, but 1098-T's for tuition payments are not available. In practice, we use the Opportunity Project Databank compiled by Census Bureau and Opportunity Insights researchers, which harmonizes these various administrative tax items over time and conveniently links filers to spouses and dependents. This linkage enables us to assign information from a mortgage 1098 that has only one spouse's tax ID number on it to both spouses in the household.[7]

The tax data only has information on home mortgages, rather than homeownership itself. To address this gap we also use property tax records, a collection of deeds and assessment data from US counties collected by Black Knight, LLC, from the year 2019. This has up to two owners listed for each residential property in most counties in the US. A small number of counties without digitized records may not be in this data, but this type of third party commercial compilation of public records is likely the most complete source of ownership data available. We use Census PVS linkages of the Black Knight data to link to the AHS sample and identify owners who own without a mortgage. 11% of the owners in the Black Knight data don't appear as an owner in the tax data. Generally speaking we are not too concerned about this because the vast majority of homeowners, especially those at a somewhat young age as in our sample, have a mortgage. This is shown in Figure 3. We use public 2019 5-year ACS estimates from IPUMS [Steven Manson and Ruggles, 2023] to show the fraction of homeowners with a mortgage by age. Given that our sample is younger (average age in 2017: 33 years old), more than 80% of owners are likely to have a mortgage.

We also use data from the FHFA, specifically their annual Housing Price Index (HPI) at the tract and county level.[8] We calculate the total percentage of home price growth by dividing the index for the county or

[7] Readers with Census Bureau-approved access should seek out the "Census CARRA Databank Manual" by Benny Goldman, November 20, 2017, for more information on which exact variables are available in Census' collection of tax data, how spouses and dependents are linked across years, and which other administrative data is used to improve said linkages, etc.

[8] FHFA Annual HPI are available from https://www.fhfa.gov/data/hpi/datasets?tab=additional-data. Bogin et al. [2019] summarizes

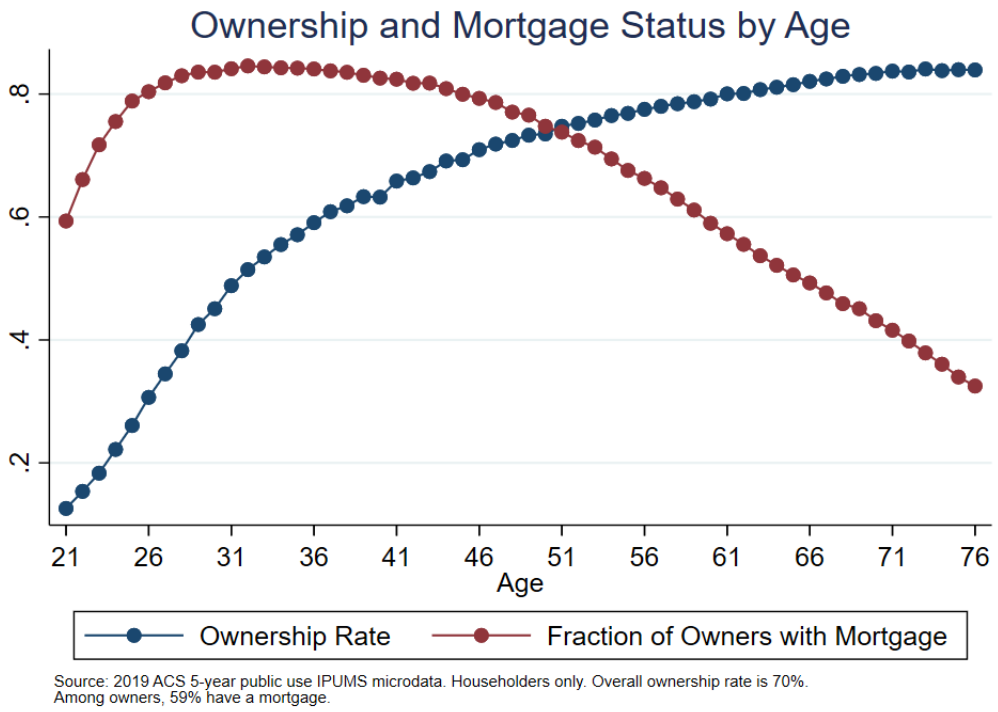


Figure 3: **Ownership and Mortgage Status by Age** Author calculations using 2019 ACS 5-year public use IPUMS microdata. Householders only. Overall ownership rate across all ages for householders is 70%. Among owners, 59% have a mortgage.

tract for 2007 divided by the value from 1995[9]. All observations can be linked to a county, while only 44% of our observations can be easily linked to a single census tract. We use tract level growth when available, and fill in observations with missing tract information with the county growth.

4 Results

4.1 Summary Statistics

The AHS is a nationally representative sample, but we are of course not able to track every respondent over the twenty-plus years of the data in our study. Over 85% of observations recieve a PIK[10]. Table 1 compares the households with PIKs to those without. As one might expect, given that in most contexts higher income individuals generally have more documentation and interact with formal administrative systems more frequently, etc., children with PIKs appear to live in homes that are better off on nearly all measures. The houses are larger and newer, with less crowding, fewer abandoned buildings nearby, and parents are more likely to own the home and be college educated, etc. Given the large sample size, nearly all of these differences are statistically significant.

In addition to being assigned a PIK through Census' PVS process, we also need respondents to file a tax return at least once between 2016 and 2019 to observe outcomes. Non-filing has become much less common in recent years, in no small part due to the Earned Income Tax Credit, which results in payouts to low-income earners that can make filing well worth their time. We find approximately 111,000 unique children with PIKs, and find tax return data in at least one year for 102,000 of them. On average the filers are slightly better off than the non-filers, as shown in columns 3 and 4 of Table 1. It is of course an unfortunate limitation to our research that we are unable to track a selected sample of survey respondents, but we believe that the low rates of attrition result in the best data available for long range tracking of respondents in the US, and that our estimates are still noteworthy.

the dataset.

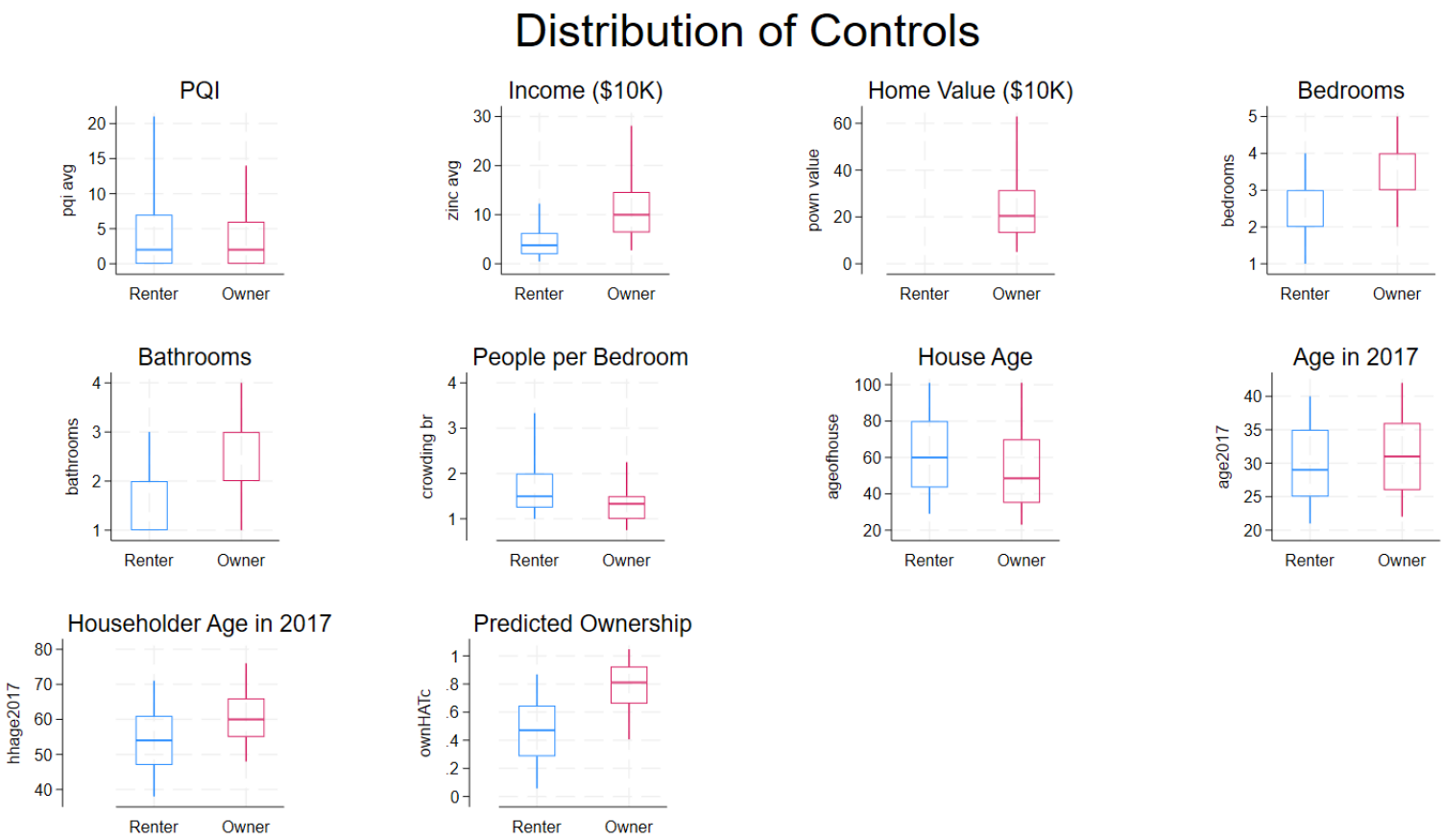
[9]Estimates are similar using a home price growth measure using the year tax records show the parents moved from the home (if any). We prefer the constant 2007-based estimate since it avoids endogeneity with the timing of potential moves.

[10]190,000 of 220,000 observations have PIKs. These numbers potentially count repeated observations of a child counting as multiple observations, since without PIKs we are not perfectly able to track individuals across rounds of the AHS survey. In most other instances in the paper we use only one observation per person.

The first two columns in Table 2 compare the children of parents who owned versus the children of those who rented. Large differences in both adult outcomes (the top section, measured with tax data) and baseline (the bottom section, measured during childhood in the AHS) are evident. For example, self-reported family income for parents who owned was $125,500, and only $47,950 for renter families. Owner families also have higher education, live in larger, newer houses, rank their neighborhood highly, and are more likely to be married. We can control for these large differences between treated (owner) and control (renter) groups, but unobservable differences are likely large as well—clearly, this is not a randomized trial.

We show the distribution of key variables using box and whisker plots for children of owners and children of renters in Figure 4. Boxes show 5th, 25th, 50th, 75th, and 95th percentiles.[11] This figure more deeply demonstrates the pattern in the summary table—children of owners grew up in higher income households with more bedrooms, more bathrooms, and slightly younger buildings. However, the distributions show significant overlap—the populations are not completely distinct. We also plot the predicted value of parental ownership based on family income, education, age, and neighborhood characteristics, and find significant overlap in the predicted value.

[11] Interpolation of 10+ observations is required for Census disclosure review to avoid disclosing exact values of medians or percentiles.



Note: Box plots show interpolated medians, 25th, and 75th percentiles, with whiskers to 5 and 95 percentiles. Variables are measured in 1995-2002 AHS. Renter/Owner refers to children of renters and children owners.

Figure 4: **Box Plots of Parental Characteristics for Renters and Owners**

We divide our linked dataset into the metro and national survey subsamples of the pooled AHS data and show a simple comparison of the two in Table 2. The metro subsample is not a panel, but is chosen from around a dozen metro areas in a particular year, with any specific metro area not repeated within our timeframe[12]. On average, metro households have lower quality housing and lower incomes, but have less crowding and are more likely to have a college degree. Of course, they are more like to live in an urban area.

4.1.1 Poor Quality Index

We use the AHS to calculate the Poor Quality Index for all households in the AHS over the last thirty years. We show summary statistics by race and ethnicity in Figure 5. The score has been decreasing for most groups (that is, housing quality has been improving). The relative ranking by race and ethnicity has changed in that Asian non-Hispanic households have surpassed white households and now have the lowest PQI. Other non-Hispanic households have had increasing PQI, though estimates seem to vary more from year to year, perhaps due to a relatively smaller sample size. Figure 6 shows the distributions of each of the seven sub-categories of PQI (electricity, heating, inside structural, bathroom, kitchen, outside structural, water and sewer). The vast majority of observations have a score of zero in any given category (between 68% to 99%), and 44% have an overall score of zero. Different regression specification such as including each sub-category of the index separately yield very similar results.

[12]See the list of metro areas chosen in a particular year at `https://www.census.gov/programs-surveys/ahs/data.html`

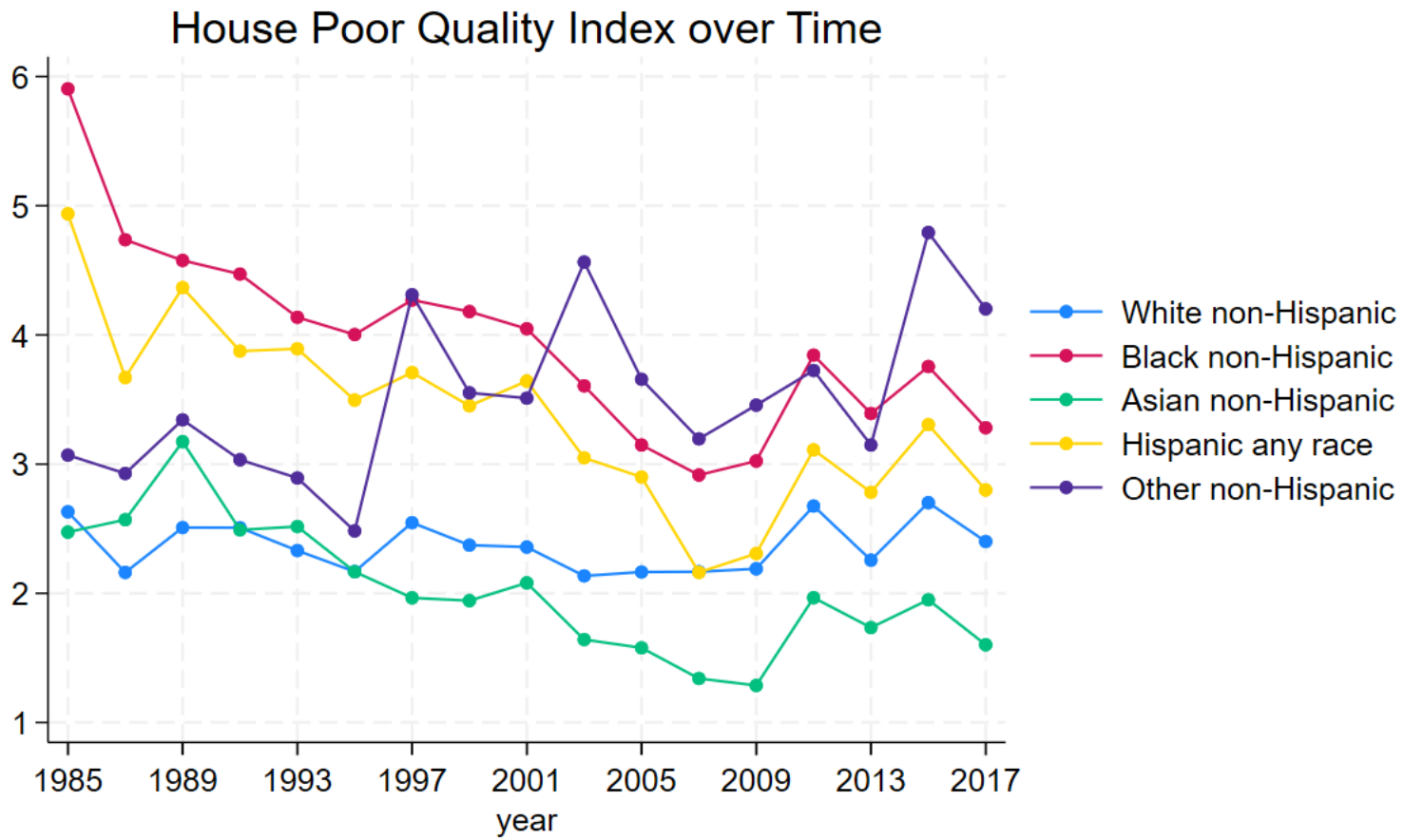


Figure 5: **Poor Quality Index Rates over Time** This chart shows the Poor Quality Index by race and ethnicity using AHS data.

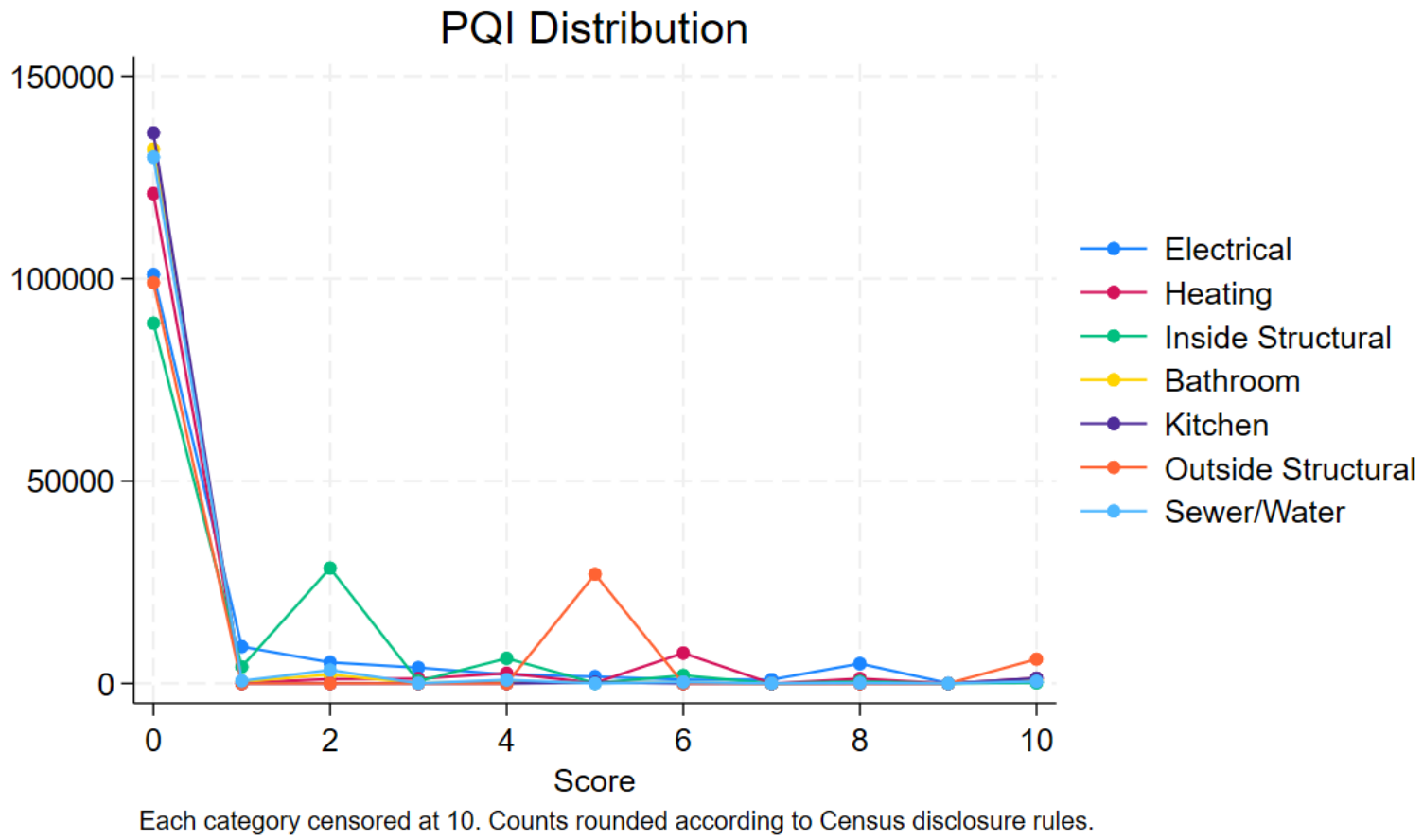


Figure 6: **Poor Quality Index Rates by Category** This chart shows the Poor Quality Index by category using 1985-2017 AHS data.

4.2 Ordinary Least Squares

Our most basic analysis is to regress recent (2016-2019) homeownership and income from W-2 and 1040 tax returns on 1995-2002 characteristics from AHS data. Controls include normalized PQI, family income, rent control, race/ethnicity, number of bedrooms, number of bathrooms, crowding, household education, whether crime is considered bothersome, age of the house, whether it was built prior to 1978, the type of home (detached single family, attached, multi-family, mobile), urban/rural status, whether there are abandoned houses nearby, whether self-reported neighborhood feel is high or low, parents marital status, sex, age of child, and metro vs. national sample indicators.

The coefficient of interest is that of parental homeownership when the subject was a child ('Parents Own' in tables). Since endogeneity is a large concern, we use the Oster [2019] method of coefficient stability and observable selection. The method uses how much the coefficient on treatment (and the R^2) change as controls are added to put bounds around the coefficient based on how much more the coefficient would change if somehow more controls were included (i.e. unobservables somehow became observable).

4.2.1 Adult Income

We find strong positive associations between growing up in a house your parents owned and later life income—parent ownership of one's childhood home is associated with $\$1,700$ to $\$2,600$ higher W2 wages as an adult. (See columns two and three of Table 3.) The first specification includes no controls, the second adds the controls listed above, and the third adds county fixed effects. A one standard deviation worse quality house, as measured by PQI, is associated with $700 lower annual wage earnings. Slightly larger differences are found using total adjusted gross income (AGI): $\$1,800$ to $\$2,800$. (See columns five and six of Table 3.) Columns seven through nine use the natural log of AGI as the outcome and find a 9 log point higher income for children of owners.

These results are robust to controlling for different versions of the PQI measure, and for whether parents own with or without a mortgage. We repeated the specifications for W2 wages using only the PQI measure of paint quality. The indicator of paint problems in the childhood home (perhaps the most likely indicator of exposure to lead paint, especially in homes built before 1978) is associated with lower W2 wages as an

adult. Additional results show similar W2 wage results but break down parental homeownership by mortgage status. After adding controls, parents having owned a home without a mortgage (i.e. owning it outright) is associated with slightly higher W2 wages than for children whose parents who owned with a mortgage.

Though these associations with wages and AGI are highly statistically significant, causality is unclear. Oster's method of partial selection correction tells us that if unobservables were equally correlated with treatment and outcome as the observables we include as controls, then the coefficients of interest would flip signs (coefficients in the 'Beta' row are negative), and that the unobservables would have to be 0.2 to 0.4 times as correlated with income and parent homeownership to drive the estimate to zero[13]. Similarly, if one treats PQI as the main treatment variable rather than parental ownership, unobservables would have to be 0.8 to 1.2 times as correlated to drive the estimate to zero–see the 'BetaPQI' and 'DeltaPQI' rows of Table 3. The method can partly be seen in action by observing how the coefficients and R^2 change between the first and third columns as additional controls are added.

Of course, it is impossible to say how correlated unobservables might be, but having controlled for income, age, and education as well as county fixed effects, we argue we have controlled for the most important measures of socioeconomic status. Therefore, it is likely that remaining unobservables are less correlated than those we already include. Still, since unobservables that are 0.5 to 0.7 times as correlated could drive some of the estimates to zero, and estimates with unobservables with equal correlation would make estimates in many specifications come out with the opposite sign, we conclude that there is not strong evidence of causality in the relationship between one's parents having owned a home and later income as an adult.

4.2.2 Other Income Statistics

We show results for four other income statistics in Table 4. For brevity we only show the specification that includes controls and fixed effects in this and all remaining regression results. Parent ownership is not statistically significantly correlated with the fraction of total income that comes from wages (column one) or with taxable dividend amounts (column two). The sign is negative for both.

Schedule E is the tax form for rental real estate and royalties as well as other income flowing through

[13]Oster's method requires an assumption regarding how much the R^2 would change with the addition of unobservable control variables. Based on data from randomized trials, Oster recommends a factor of 1.3, which is what we use throughout.

pass-through entities like partnerships. There is a small but statistically significantly higher rate of children of homeowners filing a Schedule E on their taxes (column three) or claiming any profit on those Schedule Es (column four). (The magnitudes are just over half a percentage point higher on a base of 11 percent for filing, and 0.7 percentage points off a base of 5 percent for profit.) Delta statistics for the Schedule E regressions are 0.16 for filing and 0.41 for profit (that is, unobservables 0.16 or 0.41 times as correlated as owning with outcome and controls would drive the coefficient on owning to zero). Together we interpret these as evidence that the relationship between parental homeownership and the source of later child income is weak, especially when compared to the relationships with the level of income. If we assume that most people buy their primary residence first before buying rental property, it makes sense that this relationship would not be strong in our data of adults mostly in their twenties and thirties.

4.2.3 Adult Homeownership

The results are slightly stronger for having a mortgage as an adult, which we show in Table 5. Growing up in a home owned by your parents is associated with significantly higher rates of homeownership as an adult. Children whose parents owned their home are approximately 6 percentage points more likely (off a base of 34% to 37%) to own their home as an adult. The figures looking at only mortgage data (columns 1 and 2) or looking at mortgage data plus the property tax record data (column 4) are quite similar. The effect is much smaller when looking only at those who own outright without a mortgage, but still strongly significant.

Partial selection correction shows that the effect would be near zero or slightly negative if unobservables are nearly as correlated with treatment and adult homeownership. In order to drive the estimate to zero, unobservables would generally have to be 0.6 times as correlated as the included controls. Unobservables with equal levels of correlation would produce coefficients in the opposite direction—slightly negative coefficients, on the order of 6 percentage points.

Columns five and six of Table 5 show the relationship between parent ownership and family outcomes: marital status and having any children. Children of owners are 1 percentage point more likely (off a base of 33%) to have ever been married, and 1 percentage point less likely to have any children (off a base of 39%). Perhaps this is some indication that growing up in an owned home leads to actions associated with long-term stable households and financial success (marriage and delayed child bearing) but the magnitudes seem fairly

small.

We have also broken out parental ownership by mortgage status (controlling for whether parents owned the house outright or had a mortgage) and find similar results. Parents having owned outright has similar coefficients as those for parents having had a mortgage, and children of those whose parents owned outright do not seem hugely influenced to own outright as adults themselves. We have also conducted robustness exercises for both the income and ownership regressions using multiple observations per person, one per year (of up to four years) of tax data outcomes. This latter method is mechanically similar to our main results, where our outcomes are typically averages of the four years of tax data outcomes.

4.2.4 Interactions

We interact our measure of parental ownership with a few variables of interest to investigate potential mechanisms. We interact PQI, home price index, local homeownership rate, parental income, building type, and race and ethnicity (one interaction variable or category per specification). Table 6 has W2 wages as the outcome, Table 7 has AGI, and Table 8 has homeownership. Table 9 has all the outcomes with race and ethnicity interactions.

The homeownership rate is constructed at the tract level using 2000 Decennial Census data. Including it as an additional control without interaction is not statistically significant. However, we find that the local ownership rate interaction is strongly significant for the two earnings measures as outcomes. The main coefficient on parental ownership becomes negative, implying that parents owning in an area where very few others own is associated with lower earnings, but owning in an area where nearly everyone owns is associated with higher earnings. The same pattern holds with homeownership as the outcome: parents owning in an area with low ownership is associated with only a slightly higher rate of young adult ownership (2 percentage point increase), while parents owning where nearly everyone owns is associated with an increase of 6 percentage points.

Coefficients for the interactions of ownership with parental income show no statistically significantly difference. That is, children of higher income owners do not see different outcomes compared to children of lower income owners after separately controlling for income, ownership, and other factors. The only other significant interaction is a slightly counterintuitive result from the home price index growth: specifically,

the tract level home price index from the peak before the financial crisis (2007) divided by the index at the beginning of the study period, constructed from publicly available tract and county level historical home price data from the FHFA HPI. When included without an interaction this shows no significant relationship with income. Results with ownership as the outcome are more significant. The HPI coefficient by itself is very small and insignificant. When the base term and the interaction are included, home price growth is associated with a small increased likelihood (.0121) for renters but negative (.0244) for owners. This implies that high home price growth is associated with a smaller increase for children of owners than for children of renters. This result would seem to contradict the idea that owners are able to pass their accumulated equity on to their children to help them with their own down payments. More careful evaluation of this home price growth measure is necessary[14].

The last column of interactions investigates hetereogeneity by building type: mobile home, apartment (any type of attached structure, so this includes townhomes), and the omitted group of single family detached homes. The coefficients on parents own * mobile home and parents own * apartment are consistently negative and significant at the 1% level. Owning a mobile home is consistently larger than the main coefficient for parents owning, implying children of mobile home owners are worse off than renters in both income and homeownership dimensions. Children of apartment owners have lower incomes than children of renters on average, but are still slightly more likely to own their own home than are children of renters (but less likely than children of single family home owners).

Race and Ethnicity Interactions Next we test race and ethnicity interactions, comparing non-Hispanic Blacks, Hispanics of any race, non-Hispanic Asians, and other non-Hispanics to non-Hispanic whites (the omitted group in the regressions) in Table 9. W2 wages, AGI, and ownership with a mortgage are the outcomes. For each of these, the first column interacts each race/ethnicity category with ownership. The second column is a triple interaction: race/ethnicity category with ownership and with the home price index growth rate, and ownership is interacted with the rate. The third column does the analog with the local homeownership rate.

Results using just race interactions (the first, fourth, and seventh columns) show that children of Black

[14]We have tested regressions with a potentially more endogenous measure of growth constructed from when the family moved out of the area, but prefer to use the measure with fixed start and end dates for calculation to avoid endogeneity concerns.

owners see statistically significantly lower earnings than white owners, and even slightly lower than Black *renters*. For wages, for example, the coefficient for parents owning is $\$2,230$, the Black coefficient is $-\$7,145$, and the parents own * Black coefficient is $-\$3,572$. The pattern is similar for AGI: children of Black owners see even lower incomes than children of Black renters.

No strong patterns emerge at the 1% significance level for the other race/ethnicity categories in terms of income, but there are significant relationships with ownership as the outcome. The coefficient for every included race category (Blacks, Hispanics, Asians, and all others) interacted with ownership is negative, and all but the other category is significant at the 1% level. This implies that compared to the children of white owners, children of nonwhite owners are far less likely on average to own themselves as a young adult. The coefficient for parents own * Black and parents own * Asian are similar in magnitude to the main parents owning coefficient, implying the difference between Black owners and Black renters, and between Asian owners and Asian renters, is small.

Tests with the triple interactions do not find statistically significant differences for owning in different rate environments for different race and ethnicity categories.

In summary, the strongest patterns emerge for single family homes in areas where most people own their home, which is perhaps unsurprising. We do not see strong patterns related to the quality of the house or family income levels. Perhaps surprisingly, however, children of Black renters are actually slightly better off than children of Black owners in terms income, and only slightly better off in terms of their own homeownership, generally consistent with research like Kermani and Wong [2024] showing minorities have lower rates of capital accumulation from homeownership. Further research should investigate the causes of these trends.

As an exploratory analysis we have run the race and ethnicity regressions as five separate subsample regressions instead of interactions. Each regression includes only the observations from the given race/ethnicity, effectively making every control variable an interaction term. Figures plotting the coefficients on the parents own, Z(PQI), and family income variable from each race category regression are shown in Appendix Figure A1. The takeaway is muted by the numerous comparisons: error bars to overlap for most estimates. Point estimates for income measures are mixed, but estimates show that children of Black owners see less of an ownership benefit from parental ownership compared to the other racial and ethnic categories.

4.3 Within-Family Design

Next we leverage the panel nature of the data to resolve some of the obvious endogeneity concerns. We focus on comparing across siblings within a given family. Unobservable traits such as innate desire to own, motivation, ability, tradition, etc. are removed from the measurement to the extent they are constant across the siblings within a given family. Since different children in the same family live in parent-owned housing for different durations and at different ages, we attempt to use these differences and the panel structure of the data to tease out the effect of parental ownership on children later in life for a subsample of the data. Chetty and Hendren [2018] used this method to great effect in highly cited and influential papers on the effects of neighborhood quality. The AHS is a structure-level panel, meaning the survey interviews whomever lives in the house, and does not track respondents who move away from the chosen structure. This means we are limited in our ability to measure when and whether a child who moved out of the sampled home moved into a rented or parent-owned home. However, we use parents' tax data to construct a simple residential history for children in the sample—we measure how long the child stayed in the AHS home. We observe from tax data when the parents moved out of the AHS home and if a new family moves into the surveyed home, but we cannot determine whether this move was to a rented or an owned home.

Our primary method of a within-family design limits the sample to first-time homeowners, includes fixed effects for the family (not the structure) and controls for things that vary between siblings: number of siblings, birth order, and age (using fixed effects for year of birth to flexibly control for the age-earnings profile). We construct versions of the family ID and fixed effects, using both the self-reported year the home was purchased and the AHS' recent mover group variables, to test regressions with both.

The primary coefficients of interest in these regressions are for 'Born in Owned,' and 'Years in Owned,' which measure whether a child was born while their parents already owned their home, and the number of years that a child lived in that home before their parents moved from that home. There is some evidence of a golden window at very young ages for certain biological and nutritional interventions (see for example Cusick and Georgieff [2016]), so we investigate in a similar sense whether owning the home at the time of birth has positive effects on children later in life. If parental homeownership has a causal impact on children based on the amount of time children spend in the owned home relative to their siblings, or if homeownership

improves child nutrition or health at birth in a way that carries over to adulthood, we expect these regressions to pick up evidence of the effects.

Neither of these methods show significant effects of being born in or living in homes one's parents owned relative to one's siblings, regardless of the outcome measured. Tables 10 and 11 each show three adult earnings outcomes and four homeownership outcomes. None of the main coefficients appear very large or practically meaningful. If anything, a few of the coefficients for being born into an owned home appear to be associated with a statistically significant 2 to 3 percentage point *decrease* in adult homeownership. (Women are generally associated with lower wages and lower likelihood of receiving a 1098 in their own name, but higher total family AGI or them or their spouse receiving a 1098, which is consistent with women marrying at younger ages, which we observe in our sample.) We have also run the analogous regressions using repeated values of the adult outcomes from each year of tax data. Again, there seems to be no significant effects on an individual relative to their siblings.

In some sense we do not find these results that surprising—to the extent that homeownership is a path to wealth accumulation that can be passed on to the next generation, it seems very unlikely that parents would share that wealth proportionally based on how long each child lived in their owned home. However, if living in an owned home leads to more stability and better outcomes at school, we would have expected benefits to show up here.

In addition to this within-family analysis, we have conducted within-family-over-time analysis, which includes controls for changes in household characteristics. Focusing on the 1997, 1999, and 2001 years of the AHS where survey questions are the most consistent over time, we control for changes in householder education, marital status, and income, as well as changes to the household—changes in PQI, in bedrooms (or crowding), and bathrooms. While these changes themselves are constant across siblings within a family, we interact them with the age of each child at the time of the change to see whether our main coefficients of interest change. Results are shown in Table 12. We find that results are still small and not statistically different from zero.

5 Conclusion

Results show strong and statistically significant correlations between growing up in a parent-owned (versus rented) home and later life outcomes such as higher incomes and a higher likelihood of having a mortgage. Partial selection correction investigates the likelihood that these results are causal and finds that unobservables with much lower levels of correlation between treatment and outcome as the many high-quality controls in the data could mean a true causal estimate would be zero. We also find that the income and ownership relationships are lower, and sometimes even negative, for children of Black owners.

We find similar evidence of a null result when investigating the causality question more thoroughly using within-family and within-family-over-time results. Siblings who are born into a home their parents own, or who spend more years living in the home before their parents move, are no more likely than their siblings to own their own home as an adult, and they exhibit no higher earnings.

The US spends a large amount of public funds to subsidize homeownership, and the causal effects of homeownership are hard to disentangle. Our sibling comparison method finds limited causal benefit, but has limitations. We are unable to measure important equity and wealth channels, and consider parents unlikely to share their accumulated wealth with children proportional to each child's tenure in a home they owned. Thus further research should seek out additional methods to disentangle equity and wealth from family and individual characteristics to determine to what extent homeownership has causal effects.

References

Anna Aizer and Janet Currie. Lead and juvenile delinquency: new evidence from linked birth, school, and juvenile detention records. *Review of Economics and Statistics*, 101(4):575–587, 2019.

Anna Aizer, Janet Currie, Peter Simon, and Patrick Vivier. Do low levels of blood lead reduce children's future test scores? *American Economic Journal: Applied Economics*, 10(1):307–341, 2018.

David Barker and Eric Miller. Homeownership and Child Welfare. *Real Estate Economics*, 37(2):279–303, 2009. ISSN 1540-6229. doi: 10.1111/j.1540-6229.2009.00243.x. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1540-6229.2009.00243.x. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-6229.2009.00243.x.

David R Barker. The evidence does not show that homeownership benefits children. *Cityscape*, pages 231–234, 2013.

Matteo Benetton, Marianna Kudlyak, and John Mondragon. Dynastic home equity. *Available at SSRN 4158773*, 2022.

Ariel Binder, Max Risch, and John Voorheis. Housing capital and intergenerational mobility in the united states. Technical Report 35256, National Bureau of Economic Research, 2026. URL https://www.nber.org/papers/w35256.

Jo Blanden and Stephen Machin. Home ownership and social mobility. 2017.

Jo Blanden, Andrew Eyles, and Stephen Machin. Intergenerational home ownership. *The Journal of Economic Inequality*, pages 1–25, 2023.

Kristian Blickle and Martin Brown. Borrowing constraints, home ownership and housing choice: Evidence from intra-family wealth transfers. *Journal of Money, Credit and Banking*, 51(2-3):539–580, 2019.

Samantha J Boch, Danielle M Taylor, Melissa L Danielson, Deena J Chisolm, and Kelly J Kelleher. 'home is where the health is': Housing quality and adult health outcomes in the survey of income and program participation. *Preventive medicine*, 132:105990, 2020.

Alexander Bogin, William Doerner, and William Larson. Local house price dynamics: New indices and stylized facts. *Real Estate Economics*, 47(2):365–398, 2019.

Glen Bramley and Noah Kofi Karley. Homeownership, Poverty and Educational Achievement: School Effects as Neighbourhood Effects. *Housing Studies*, 22(5):693–721, September 2007. ISSN 0267-3037. doi: 10.1080/02673030701474644. URL https://doi.org/10.1080/02673030701474644. Publisher: Routledge _eprint: https://doi.org/10.1080/02673030701474644.

Kerwin Kofi Charles and Erik Hurst. The transition to home ownership and the black-white wealth gap. *Review of Economics and Statistics*, 84(2):281–297, 2002.

Raj Chetty and Nathaniel Hendren. The impacts of neighborhoods on intergenerational mobility i: Childhood exposure effects. *The Quarterly Journal of Economics*, 133(3):1107–1162, 2018.

Sarah E Cusick and Michael K Georgieff. The role of nutrition in brain development: the golden opportunity of the "first 1000 days". *The Journal of pediatrics*, 175:16–21, 2016.

Rebecca Diamond and William F Diamond. Racial differences in the total rate of return on owner-occupied housing. Technical report, National Bureau of Economic Research, 2024. URL https://www.nber.org/papers/w32916.

Paul Diegert, Matthew A Masten, and Alexandre Poirier. Assessing omitted variable bias when the controls are endogenous. *arXiv preprint arXiv:2206.02303*, 2022.

Frederick Eggers and Fouad Moumen. American housing survey: Housing adequacy and quality as measured by the ahs. *Available at SSRN 2284174*, 2013.

Peter Englund, Thomas Jansson, and Todd Sinai. How parents influence the wealth accumulation of their children. Technical report, Working Paper University of Pennsylvania, 2014.

Government Accountability Office. K-12 education: Many challenges arise in educating students who change schools frequently. report to congressional requesters. gao-11-40. *US Government Accountability Office*, 2010.

Richard K Green. Do kids of homeowners do better than kids of renters? *Cityscape*, pages 223–225, 2013.

Richard K. Green and Michelle J. White. Measuring the Benefits of Homeowning: Effects on Children. *Journal of Urban Economics*, 41(3):441–461, May 1997. ISSN 0094-1190. doi: 10.1006/juec.1996.2010. URL https://www.sciencedirect.com/science/article/pii/S0094119096920100.

Luigi Guiso and Tullio Jappelli. Private transfers, borrowing constraints and the timing of homeownership. *Journal of Money, Credit and Banking*, pages 315–339, 2002.

Emily W Harville and Felicia A Rabito. Housing conditions and birth outcomes: the national child development study. *Environmental research*, 161:153–157, 2018.

Donald Haurin. The relationship of homeownership, house prices, and child well-being. *Cityscape*, pages 227–229, 2013.

Donald R. Haurin, Toby L. Parcel, and R. Jean Haurin. Does Homeownership Affect Child Outcomes? *Real Estate Economics*, 30(4):635–666, 2002. ISSN 1540-6229. doi: 10.1111/1540-6229.t01-2-00053. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/1540-6229.t01-2-00053. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/1540-6229.t01-2-00053.

David E Jacobs, Robert P Clickner, Joey Y Zhou, Susan M Viet, David A Marker, John W Rogers, Darryl C Zeldin, Pamela Broene, and Warren Friedman. The prevalence of lead-based paint hazards in us housing. *Environmental health perspectives*, 110(10):A599–A606, 2002.

Amir Kermani and Francis Wong. Racial disparities in housing returns. Technical report, National Bureau of Economic Research, September 2024.

Lisa L. Mohanty and Lakshmi K. Raut. Home Ownership and School Outcomes of Children: Evidence from the PSID Child Development Supplement. *The American Journal of Economics and Sociology*, 68(2):465–489, 2009. ISSN 1536-7150. doi: 10.1111/j.1536-7150.2009.00635.x. URL https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1536-7150.2009.00635.x. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1536-7150.2009.00635.x.

Sandra J Newman and C Scott Holupka. Looking back to move forward in homeownership research. *Cityscape*, pages 235–246, 2013.

Emily Oster. Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204, 2019.

Amy Ellen Schwartz, Leanna Stiefel, and Sarah A Cordes. Moving matters: The causal effect of moving schools on student performance. *Education Finance and Policy*, 12(4):419–446, 2017.

David Van Riper Katherine Knowles Tracy Kugler Finn Roberts Steven Manson, Jonathan Schroeder and Steven Ruggles. Ipums national historical geographic information system: Version 18.0 [dataset], 2023.

Table 1: Summary Statistics: PIK, no-PIK & Tax Filers, Non-Filers

Variable	Has PIK	No PIK	IRS Filed	Never Filed
PQI	4.08	3.98	4.42	5.17
Family Income	$66,370	$51,900	$66,980	$50,630
Parents Owned Home	0.71	0.51	0.7	0.56
White non-Hispanic	0.68	0.5	0.68	0.54
Bedrooms	3.2	2.95	3.25	3.07
Bathrooms	2.04	1.86	2.09	1.84
People per Bedroom	1.46	1.69	1.46	1.54
College Degree	0.18	0.14	0.18	0.11
Urban	0.8	0.84	0.81	0.84
House Age	56.28	58.22	55.61	59.29
Built pre-1978	0.67	0.7	0.66	0.73
Single Family Detached	0.75	0.6	0.75	0.65
Abandonded Nearby	0.05	0.08	0.06	0.1
Neighborhood Rank High	0.41	0.36	0.39	0.35
Parents Married	0.73	0.65	0.7	0.55
Age in 2017	30.82	30.48	32.6	33.64
Female	0.49	0.48	0.5	0.34
N	190,000*	30,000*	102,000	8,700

Table shows average characteristics of those in the AHS with and without PIKs (with potentially more than one observation per person), and among those with PIKs (only one observation per person) those who filed taxes and those who did not.

Table 2: Summary Stats: Owners/Renters and Metro/National

Variable	Own	Rent	Metro	National
W2 Wages	47,940	33,430	44,820	40,720
1040 AGI	79,700	50,160	73,130	66,140
1098-Owner (Self)	.26	.14	.23	.21
1098-Owner (Couple)	.4	.32		
Wage/Total Income Ratio	.81	1.48	1.14	.83
Taxable Dividends	512	134.4	398.5	381
Schedule A [0/1]	.34	.19	.31	.27
Schedule C [0/1]	.29	.26	.28	.27
Schedule E [0/1]	.13	.06	.11	.10
Schedle E Any Profit	.06	.03	.05	.05
Taxable Interest	206.6	57.07	169.6	141.8
Ever Married	.37	.25	.33	.32
Any Children	.39	.38	.38	.39
Poor Quality Index	3.97	5.44	5.1	3.44
Family Income	125,500	47,950	76,380	88,950
Parents Owned Home	1	0	.7	.63
White non-Hispanic	.76	.47	.67	.65
Bedrooms	3.52	2.55	3.23	3.16
Bathrooms	2.33	1.5	2.12	1.96
People per Bedroom	1.36	1.76	1.42	1.61
College Degree	.22	.09	.2	.15
Urban	.78	.9	.86	.76
House Age	52.9	61.16	53.59	58.84
Built pre-1978	.6	.76	.62	.71
Single Family Detached	.91	.35	.75	.68
Abandoned Houses Nearby	.04	.11	.05	.09
Neighborhood Rank High	.43	.29	.4	.36
Parents Married	.82	.45	.72	.66
Age in 2017	31.01	29.99	30.5	30.97
Female	.49	.5	.5	.49
N	92,500	45,500	84,500	53,000

Table shows average characteristics of those in the AHS whose parents owned houses vs. rented, and the National vs. Metro AHS samples. Top rows are recent tax data outcomes, lower are childhood AHS data.

Table 3: Wages and Total Income

Outcome	(1) Basic W2 Wages	(2) Full Controls W2 Wages	(3) County FE W2 Wages	(4) Basic AGI	(5) Full Controls AGI	(6) County FE AGI	(7) Basic ln(AGI)	(8) Full Controls ln(AGI)	(9) County FE ln(AGI)
Z(PQI)	–1,479***	–641.4***	–709.7***	–2,731***	–909.2***	–1,076***	-.0461***	-.0238***	-.0262***
	[228.1]	[173.5]	[165.1]	[466.7]	[331.9]	[333.2]	[0.00540]	[0.00334]	[0.00327]
Family Income		78.84**	57.97**		257.4**	236**		.00144**	.00118**
		[34.20]	[27.30]		[110.1]	[106.8]		[0.000593]	[0.000514]
Parents Own	13,490***	2,564***	1,707***	27,390***	2,759***	1,832*	.407***	.0979***	.0878***
	[878]	[425.1]	[438.1]	[1,884]	[1,048]	[1,032]	[0.0226]	[0.0103]	[0.00936]
Observations	112,000	112,000	112,000	113,000	113,000	113,000	112,000	112,000	112,000
R-squared	.012	.098	.106	.007	.061	.065	.036	.251	.263
Mean	42,960	42,960	42,960	69,510	69,510	69,510	10.65	10.65	10.65
Controls	NO	YES	YES	NO	YES	YES	NO	YES	YES
CountyFE	NO	NO	YES	NO	NO	YES	NO	NO	YES
Beta		–6562	–8754		–16830	–20010		-.163	-.202
Delta		.343	.21		.176	.109		.445	.374
BetaPQI		–81.34	–134.9		249	104.1		-.0083	-.0104
DeltaPQI		1.137	1.216		.793	.917		1.495	1.586

Robust standard errors clustered at county level in brackets. Columns one to three use W2 wages as the outcome, and columns four to six use AGI. Beta and Delta are Oster (2019) statistics calculated using the treatment coefficient Parents Own.

Table 4: Investment Income

Outcome	(12) County FE Wage Income Ratio	(15) County FE Dividends	(18) County FE Schedule E	(21) County FE Sched E Profit
Z(PQI)	-.205	–1.904	-.000849	-.000482
	[0.196]	[25.13]	[.001]	[.0006]
Family Income	.00352	18.31**	.000478***	.000159**
	[0.00395]	[8.270]	[.0002]	[.0001]
Parents Own	-.472	–63.88	.00596**	.00655***
	[0.488]	[90.04]	[.003]	[.002]
Observations	107,000	116,000	116,000	123,000
R-squared	.003	.012	.082	.044
Mean	1.044	385	.106	.0494
Controls	YES	YES	YES	YES
CountyFE	YES	YES	YES	YES
Beta	-.226	–431.4	-.0426	-.0135
Delta	1.478	-.26	.158	.407
BetaPQI	-.276	8.189	.0023	.001
DeltaPQI	–3.76	.199	.286	.346

Robust standard errors clustered at county level in brackets. Column one outcome is W2 wages/AGI. Column two is dividend income, three is an indicator of filing schedule E, and four is an indicator for positive profit on schedue E. Beta and Delta are Oster (2019) statistics calculated using the treatment coefficient Parents Own, and Z(PQI).

Table 5: Home Ownership and Family Status

Outcome	(24) County FE Mortgage (Self)	(27) County FE Mortgage (Couple)	(30) County FE Own (No Mort)	(33) County FE Own (Any)	(36) County FE Married	(39) County FE Any Children
Z(PQI)	-.00548***	-.00749***	-.000287	-.00778***	-.00232*	.00161
	[.0011]	[.0012]	[.0006]	[.0013]	[.0013]	[.0016]
Family Income	.00002	.00003	.00004	.00007	-.00007	-.00006
	[9.56e-05]	[9.55e-05]	[4.03e-05]	[0.000103]	[6.25e-05]	[6.75e-05]
Parents Own	.0465***	.0558***	.00664***	.0624***	.0112***	-.0122***
	[.0037]	[.004]	[.0014]	[.0044]	[.0041]	[.004]
Observations	123,000	123,000	123,000	123,000	123,000	123,000
R-squared	.137	.207	.013	.208	.193	.159
Mean	.22	.342	.0298	.372	.328	.389
Controls	YES	YES	YES	YES	YES	YES
CountyFE	YES	YES	YES	YES	YES	YES
Beta	-.026	-.0622	.00651	-.0574	-.0808	-.0296
Delta	.723	.559	2.841	.608	.161	–1.525
BetaPQI	-.00182	-.00149	-.0000214	-.00151	.0018	.000181
DeltaPQI	1.45	1.228	1.075	1.221	.58	1.117

Robust standard errors clustered at county level in brackets. Beta and Delta are Oster (2019) statistics calculated using the treatment coefficient Parents Own.

Table 6: Interactions: W2 Wages as Outcome

Outcome	(1) W2 Wages	(3) W2 Wages	(4) W2 Wages	(6) W2 Wages	(7) W2 Wages	(9) W2 Wages	(10) W2 Wages
Parents Own	1,681***	1,756***	2,428**	1,641***	–5,919***	1,369**	3,005***
	[438.7]	[443]	[1,226]	[426.8]	[2,080]	[606.4]	[474.4]
Family Income	58.04**	57.76**	57.98**	57.51**	56.78**	5.311	57.46**
	[27.30]	[27.48]	[27.46]	[27.15]	[27.08]	[71.39]	[27.16]
Home Quality	–828.6***	–701.2***	–700.3***	–709.7***	–715.6***	–713.5***	–705.9***
	[191.5]	[167.8]	[167.5]	[165.2]	[165.5]	[164.7]	[164.1]
	PQI	HPI	HPI	OwnRate	OwnRate	Income	BuildingType
Main Interaction	257		–330.9		11,880***	53.73	
	[250.6]		[577.6]		[3,140]	[73.23]	
Home Price Growth		673.1	844.8				
		[438.5]	[576.7]				
Home Ownership Rate				2,661	–3,810		
				[2,195]	[2,972]		
Mobile Home							1,749
							[1,078]
Apartment							2,008***
							[661.4]
Parents Own * Mobile Home							–6,355***
							[1,242]
Parents Own * Apartment							–4,175***
							[1,142]
Observations	112,000	110,000	110,000	112,000	112,000	112,000	112,000
R-squared	.106	.106	.106	.106	.107	.106	.107
Controls	YES	YES	YES	YES	YES	YES	YES
CountyFE	YES	YES	YES	YES	YES	YES	YES
Mean	42,960	43,080	42,960	42,960	42,960	42,960	42,960

Robust standard errors clustered at county level in brackets. One interaction variable per column, with the coefficients in the same row.

Table 7: Interactions: AGI as Outcome

Outcome	(11) AGI	(13) AGI	(14) AGI	(16) AGI	(17) AGI	(19) AGI	(20) AGI
Parents Own	1,783*	1,693*	205.8	1,719*	–13,390***	691.2	3,561***
	[1,058]	[1,027]	[2,267]	[1,041]	[4,632]	[1,917]	[975.1]
Family Income	236.1**	236.1**	235.7**	235.2**	233.9**	59.43	235.3**
	[106.8]	[106.8]	[107.1]	[106.3]	[106.4]	[212]	[106.6]
Home Quality	–1,315***	–1,001***	–1,003***	–1,076***	–1,088***	–1,089***	–1,073***
	[317.7]	[330.5]	[330.4]	[332.6]	[333.4]	[329.3]	[333.3]
	PQI	HPI	HPI	OwnRate	OwnRate	Income	BuildingType
Main Interaction	502.9		730.7		23,730***	179.9	
	[742.7]		[1,113]		[6,541]	[241.9]	
Home Price Growth		–313.6	–705.3				
		[1,184]	[1,422]				
Home Ownership Rate				4,542	–8,618		
				[5,040]	[6,236]		
Mobile Home							3,907**
							[1,837]
Apartment							5,612***
							[1,099]
Parents Own * Mobile Home							–9,842***
							[2,085]
Parents Own * Apartment							–5,275**
							[2,419]
Observations	113,000	111,000	111,000	113,000	113,000	113,000	113,000
R-squared	.065	.065	.065	.065	.065	.065	.065
Controls	YES	YES	YES	YES	YES	YES	YES
CountyFE	YES	YES	YES	YES	YES	YES	YES
Mean	69,510	69,660	69,510	69,510	69,510	69,510	69,510

Robust standard errors clustered at county level in brackets. One interaction variable per column, with the coefficients in the same row.

Table 8: Interactions: Homeownership as Outcome

Outcome	(21) Mortgage (Couple)	(23) Mortgage (Couple)	(24) Mortgage (Couple)	(26) Mortgage (Couple)	(27) Mortgage (Couple)	(29) Mortgage (Couple)	(30) Mortgage (Couple)
Parents Own	.0557***	.056***	.106***	.0554***	.0177	.0545***	.0653***
	[0.00407]	[0.00406]	[0.00992]	[0.00406]	[0.0142]	[0.00523]	[0.00449]
Family Income	.0000345	.000033	.0000485	.000032	.0000281	-.000155	.0000302
	[9.56e-05]	[9.54e-05]	[9.66e-05]	[9.52e-05]	[9.56e-05]	[0.000504]	[9.52e-05]
Home Quality	-.00758***	-.00733***	-.00725***	-.0075***	-.00753***	-.00751***	-.00748***
	[0.00132]	[0.00118]	[0.00118]	[0.00117]	[0.00117]	[0.00117]	[0.00117]
	PQI	HPI	HPI	OwnRate	OwnRate	Income	BuildingType
Main Interaction	.00018		-.0244***		.0592***	.000194	
	[0.00232]		[0.00434]		[0.0198]	[0.000519]	
Home Price Growth		-.000739	.0121***				
		[0.00328]	[0.00315]				
Home Ownership Rate				.0152	-.0168		
				[0.0115]	[0.0151]		
Mobile Home							-.0398**
							[0.0161]
Apartment							.0176***
							[0.00622]
Parents Own * Mobile Home							-.0606***
							[0.0186]
Parents Own * Apartment							-.0279***
							[0.00956]
Observations	123,000	122,000	122,000	123,000	123,000	123,000	123,000
R-squared	.207	.206	.207	.207	.207	.207	.207
Controls	YES	YES	YES	YES	YES	YES	YES
CountyFE	YES	YES	YES	YES	YES	YES	YES
Mean	.342	.342	.342	.342	.342	.342	.342

Robust standard errors clustered at county level in brackets. One interaction variable per column, with the coefficients in the same row.

Table 9: Interactions: Race and Ethnicity

Outcome	(2) Race W2 Wages	(5) HPI W2 Wages	(8) Ownership Rate W2 Wages	(12) Race AGI	(15) HPI AGI	(18) Ownership Rate AGI	(22) Race Mortgage (Couple)	(25) HPI Mortgage (Couple)	(28) Ownership Rate Mortgage (Couple)
Parents Own	2,230***	2,881	–8,354*	2,621**	2,032	–19,580**	.0743***	.118***	.0616**
	[517.1]	[2,142]	[4,516]	[1,256]	[4,451]	[8,898]	[0.00548]	[0.0167]	[0.0286]
Family Income	56.46**	54.85*	53.28*	232.8**	234.7**	230.2**	.000024	.0000379	.0000187
	[27.46]	[28.30]	[28.65]	[106.9]	[106]	[106.9]	[9.58e-05]	[9.95e-05]	[9.67e-05]
Home Quality	–722***	–713.7***	–728.6***	−1,115***	−1,031***	−1,112***	−.00756***	−.00729***	−.00755***
	[164.7]	[165.8]	[165.7]	[327]	[324.4]	[327.1]	[0.00116]	[0.00118]	[0.00116]
Rate (HPI/Ownership)		1,222	–8,740		–518.1	–19,310		-.00315	.0311
		[1,039]	[6,213]		[2,472]	[12,170]		[0.00634]	[0.0332]
Own * Rate		–270.3	16,060**		262.6	33,650***		-.0223***	.016
		[1,094]	[6,568]		[2,257]	[12,720]		[0.00806]	[0.0394]
Black	–7,145***	−5,081**	−12,100**	−21,930***	−21,520***	−30,710***	−.168***	−.212***	−.135***
	[1,016]	[2,168]	[4,835]	[1,799]	[5,234]	[8,717]	[0.00774]	[0.0149]	[0.0286]
Hispanic	−979.3	−1,587	−8,676**	−7,402***	−9,263	−25,730***	−.0335***	−.0699***	-.0268
	[884.5]	[2,655]	[4,357]	[1,690]	[5,766]	[8,841]	[0.00731]	[0.0232]	[0.0313]
Asian	8,109***	10,050**	435.9	4,506*	11,370	−13,360	.00547	−.105***	.0536
	[1,271]	[5,096]	[5,216]	[2,642]	[8,547]	[10,540]	[0.0146]	[0.0305]	[0.0329]
Other	–2,910*	–8,631**	–6,201	–3,462	−20,640**	−13,440	-.0715***	-.186***	.025
	[1,693]	[3,967]	[6,068]	[4,158]	[8,018]	[13,150]	[0.0192]	[0.0411]	[0.0416]
Parents Own * Black	–3,572***	–2,460	5,333	–9,071***	–9,131	8,285	-.0616***	-.0821***	-.0539
	[993.3]	[2,659]	[5,409]	[2,091]	[6,129]	[11,080]	[0.00975]	[0.0217]	[0.0409]
Parents Own * Hispanic	–1,412*	2,205	8,480	–1,665	–3,802	19,440	-.0294***	-.0316	-.00637
	[745.7]	[2,968]	[5,197]	[2,860]	[6,762]	[12,320]	[0.00913]	[0.0328]	[0.0448]
Parents Own * Asian	1,846	7,024	7,133	10,310**	8,942	14,570	-.0675***	-.0236	-.0598
	[2,107]	[7,845]	[9,540]	[4,056]	[13,260]	[21,690]	[0.0126]	[0.0441]	[0.0559]
Parents Own * Other	13,280	3,090	–38,560*	26,230	69,380	9,585	-.00516	.00151	-.102
	[8,232]	[8,627]	[22,420]	[16,440]	[42,640]	[51,250]	[0.0233]	[0.0665]	[0.102]
Rate * Black		–1,051	7,425		–169.2	12,420		.0235***	-.0604
		[1,296]	[7,024]		[2,261]	[12,640]		[0.00687]	[0.0413]
Rate * Hispanic		196.5	12,490**		982.1	30,230**		.0146	-.00917
		[1,310]	[6,362]		[2,392]	[13,260]		[0.00948]	[0.0484]
Rate * Asian		–806.7	12,490		–2,482	29,380*		.042***	-.0865
		[1,866]	[8,593]		[3,107]	[16,870]		[0.0126]	[0.0545]
Rate * Other		2,802	4,786		8,129**	15,210		.0568***	-.167**
		[2,054]	[9,408]		[3,578]	[22,450]		[0.0196]	[0.0703]
Rate * Parents Own * Black		–708.4	–13,060		–34.85	–24,710		.0105	-.00156
		[1,590]	[8,037]		[2,755]	[15,660]		[0.0100]	[0.0597]
Rate * Parents Own * Hispanic		–1,506	–15,430*		787.7	–33,500*		.00528	-.0344
		[1,330]	[8,232]		[3,541]	[17,490]		[0.0131]	[0.0666]
Rate * Parents Own * Asian		–2,115	–8,389		304.1	–7,639		-.011	.00307
		[2,819]	[14,200]		[5,236]	[31,260]		[0.0168]	[0.0840]
Rate * Parents Own * Other		4,824	77,860*		–19,740	25,090		-.00231	.168
		[5,421]	[42,070]		[14,410]	[68,940]		[0.0301]	[0.156]
Observations	112,000	110,000	112,000	113,000	111,000	113,000	123,000	122,000	123,000
R-squared	.107	.106	.107	.065	.065	.066	.208	.208	.208
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
CountyFE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Mean	42,960	42,960	42,960	69,510	69,510	69,510	.342	.342	.342

Robust standard errors clustered at county level in brackets. For each outcome (W2 wages, AGI, and homeownership), the first column interacts ownership with each race/ethnicity category. The second includes and interacts with the Home Price Index growth rate and the third includes and interacts with the local homeownership rate.

Table 10: Fixed Effect Regressions of Birth into Home Parents Owned

	(1)	(6)	(11)	(16)	(21)	(26)	(31)
OUTCOME	Wages	AGI	ln(AGI)	1098	1098 Couple	Own No Mortgage	Own Any
Born in Owned	–481.5	1,125	–0.0685	–0.0269**	–0.0282**	0.00665	–0.0216*
	[2,028]	[4,839]	[0.0464]	[0.0121]	[0.0127]	[0.00493]	[0.0127]
Female	–15,270***	10,430***	0.129***	–0.173***	0.0469***	–0.00267	0.0443***
	[1,057]	[3,333]	[0.0236]	[0.00617]	[0.00639]	[0.00250]	[0.00638]
Older Siblings	YES	YES	YES	YES	YES	YES	YES
Family ID FE	YES	YES	YES	YES	YES	YES	YES
Birth Year FE	YES	YES	YES	YES	YES	YES	YES
Birth Order FE	YES	YES	YES	YES	YES	YES	YES
Observations	27,500	30,000	29,500	31,500	31,500	31,500	31,500

Notes: Robust standard errors in brackets. Sample includes first time home-buyers only.

Table 11: Fixed Effect Regressions of Years in Owned Home

	(5)	(10)	(15)	(20)	(25)	(30)	(35)
OUTCOME	Wages	AGI	ln(AGI)	1098	1098 Couple	Own No Mortgage	Own Any
Years in Owned	–70.08	–278.4	0.00122	–0.000597	0.00145	0.000207	0.00166
	[275]	[790.5]	[0.00883]	[0.00190]	[0.00195]	[0.000836]	[0.00192]
Female	–15260***	10430***	0.13***	–0.173***	0.0471***	–0.0027	0.0444***
	[1,060]	[3,332]	[0.0236]	[0.00618]	[0.00639]	[0.00250]	[0.00638]
Older Siblings	YES	YES	YES	YES	YES	YES	YES
Family ID FE	YES	YES	YES	YES	YES	YES	YES
Birth Year FE	YES	YES	YES	YES	YES	YES	YES
Birth Order FE	YES	YES	YES	YES	YES	YES	YES
Observations	27,500	30,000	29,500	31,500	31,500	31,500	31,500

Robust standard errors in brackets. Years in owned defined using child birthdate and parent ZIP code in tax data.

Table 12: Time Varying Characteristics of Parents Interacted with Child Age

	(4)	(12)	(20)	(28)	(36)	(44)	(52)
OUTCOME	Wages	AGI	ln(AGI)	1098	1098 Couple	Own No Mortgage	Own Any
Years in Owned	–227.8	45.39	0.00783	0.00209	0.0056	0.000711	0.00631
	[556.6]	[1,095]	[0.0136]	[0.00389]	[0.00399]	[0.00153]	[0.00405]
Female	YES	YES	YES	YES	YES	YES	YES
Differences* Age at Move-In	YES	YES	YES	YES	YES	YES	YES
Family ID FE	YES	YES	YES	YES	YES	YES	YES
Birth Year FE	YES	YES	YES	YES	YES	YES	YES
Birth Order FE	YES	YES	YES	YES	YES	YES	YES
Observations	5,900	6,300	6,300	6,800	6,800	6,800	6,800

Robust standard errors in brackets. Regressions include fixed effects, as well as controls for changes in home ownership, income, PQI, bedrooms, bathrooms, education, crowding, and marital status, each interacted with the age of the child at the time parents purchased home. Sample limited to those with repeated observations 1997, 1999, and 2001.

A Additional Results

We have conducted several additional robustness checks that have been disclosed for earlier preliminary versions of the paper. Results are generally similar to those included in the paper.

These additional specifications include:

- Breaking out the home quality meausure (PQI) to focus only on the paint quality aspect and interacting that with pre-1978 construction (an indicator of possible lead paint exposure)
- Breaking out parental ownership control into owning outright and owning with a mortgage
- The combination of the above (Paint problem, Pre-1978 construction * Paint problem, parents own outright, parents own with a mortgage controls)
- Testing different types of child ownership (with and without mortgage) combined with different types of parent ownership (with and without mortgage).
- Repeating some of the main specifications with one observation per year of outcome data (2016-2019) and adding fixed effects for each year. The versions in the main body of the paper use an average of all observed years of outcome data as a single observation, which mechanically produces similar results.
- Repeating the fixed effect sibling comparison regression with one observation per year of outcome data.

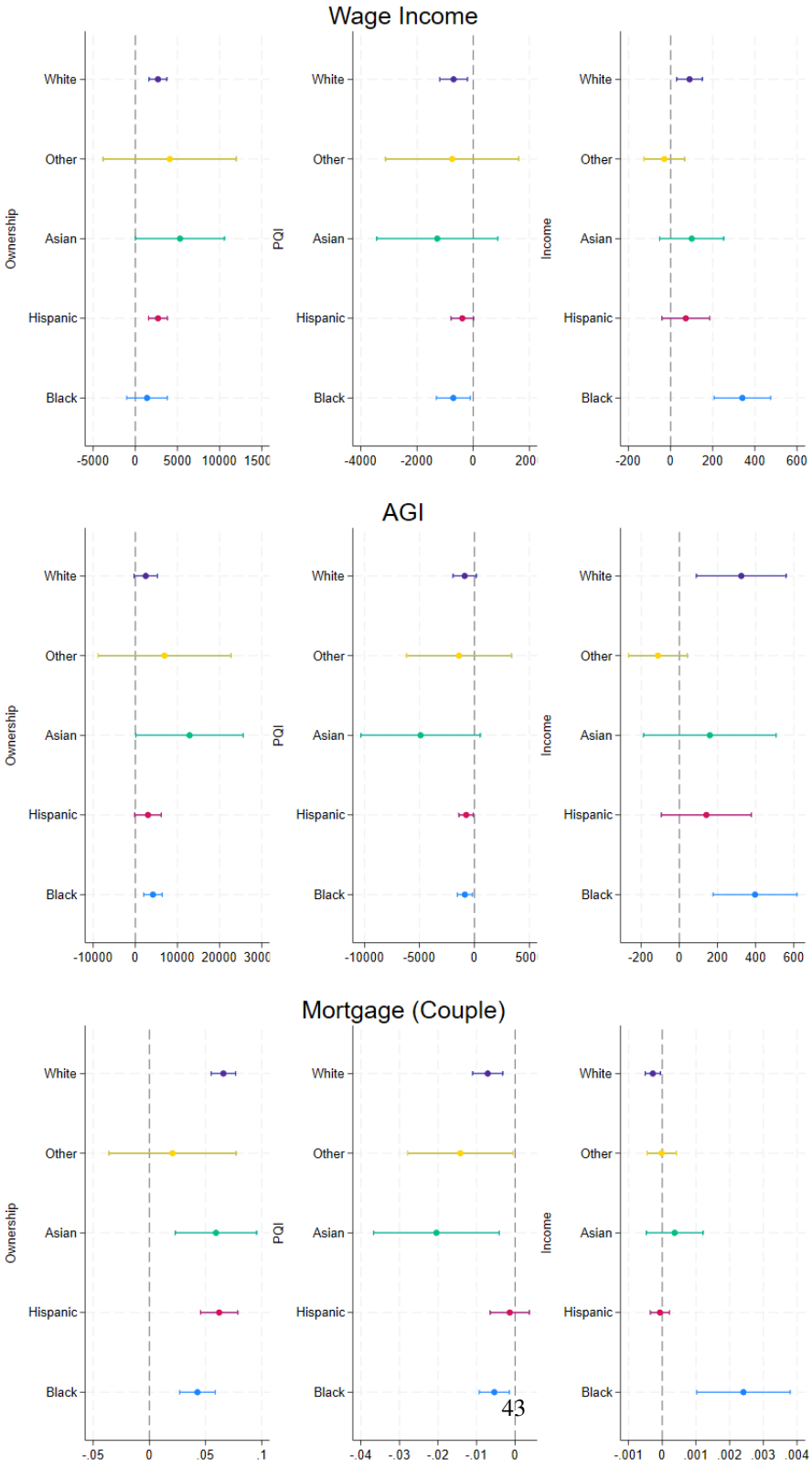


Figure A1: **Coefficients on regressions of each race/ethnicity group separately. W2 Wages, AGI, and Homeownership outcomes, with parental ownership, PQI, and family income as controls.**